

MOHAMED HANY REDA

Full-Stack Software Engineer

Remote — available worldwide, open to relocation | 3+ years of experience | led engineering teams of up to 10

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PROFESSIONAL SUMMARY

Full-stack software engineer with 3+ years shipping production web, mobile and AI-enabled products end to end — architecture, backend APIs, frontend UI, automated testing and deployment. Sole owner of a multi-tenant ERP built on FastAPI, PostgreSQL and React with database-level tenant isolation, and of a live e-commerce platform with zero-downtime releases. Comfortable taking a feature from requirements through production, and have also led engineering teams of up to 10 and owned project budgets when a project needed it. Native Arabic, C2 English, B2 Russian — able to work directly with teams across three markets.

TECHNICAL SKILLS

Frontend: JavaScript, React, Three.js, Flutter/Dart, responsive UI, cross-browser development

Backend: Python, FastAPI, Node.js, Firebase Cloud Functions, REST APIs, Clean Architecture, RBAC, multi-tenant systems

Databases: PostgreSQL, SQLAlchemy, Redis, SQLite, Row-Level Security

DevOps & Tooling: Docker, Git, CI, Pytest, Ruff, MyPy

Delivery & Leadership: Agile, Scrum, Kanban, requirements analysis, SRS, user stories, backlog, risk and release management; team leadership up to 10, budget ownership, stakeholder communication

Also experienced with: PyTorch, computer vision, reinforcement learning (PPO/SAC/A2C), Grad-CAM explainability, Unreal Engine 5

KEY PROJECTS

HMS — Multi-Tenant Enterprise ERP & AI Platform

Python · FastAPI · PostgreSQL · SQLAlchemy · Redis · React · Docker — [repository](#)

- Sole owner of a multi-tenant ERP covering inventory, procurement, sales/CRM, finance, maintenance and HR in one Clean Architecture codebase.
- Enforced tenant isolation with PostgreSQL Row-Level Security at the database layer rather than trusting application code.
- Saudi ZATCA Phase 1 e-invoicing compliance, RBAC and audit logging, full Arabic/English RTL support.

AuraGift — Live E-Commerce Platform

JavaScript · Node.js · Firebase Cloud Functions · Firebase Hosting · UX/UI — [auragift.web.app](#)

- Own the client side and UX of a premium digital-gifting platform in production.
- Built serverless order- and payment-processing logic with Node.js on Firebase Cloud Functions.
- Set up and maintain a repeatable, zero-downtime deployment pipeline on Firebase Hosting.

High-Throughput Sorting-Center Digital Twin

Python · PyTorch · Three.js · Discrete-Event Simulation · Pytest — [repository](#)

- Built a 400-lane logistics digital twin on a custom deterministic simulation engine, modelling ~96,400 items/hour.
- Browser-based 3D operations view (Three.js) and a full automated test suite of 530+ tests.

Fracture Detection AI — Explainable Medical Imaging

Python · PyTorch · EfficientNet-B0 · Streamlit — [repository](#)

- End-to-end computer-vision product for bone-fracture detection: 90.21% accuracy, Grad-CAM explanations.
- Shipped as a Streamlit web app and Telegram bot with automated PDF reporting.

Research and portfolio software. Not a certified medical device.

Additional engineering projects

Reinforcement-learning research, multi-agent simulation, cross-platform apps, Unreal Engine

- Full list and source code at [hany15.github.io/Hany-Reda-Portfolio](#).

PROFESSIONAL EXPERIENCE

Full-Stack Developer / Technical Owner | HMS — medical equipment manufacturer

Jan 2024 — Present

- Own the corporate web product end to end: requirements, architecture, front end and back end, release and support.
- Make the technical and architectural calls; integrate with internal company systems; surface technical risk before release.

Full-Stack Developer / Product Delivery Owner | AuraGift — digital gifting e-commerce

Jan 2024 — Present

- Own the client side and UX of a live premium platform, from requirement through QA to production deploy.
- Built serverless Node.js functions on Firebase Cloud Functions to process orders and payments.
- Built the interface and UX layer — responsive layout, animation, cross-browser behaviour, performance tuning.

Technical Project Lead / Full-Stack Developer | HMS

Jan 2023 — Jun 2023

- Led a team of 8 on a corporate web product while contributing to the build as a full-stack developer.
- Independently managed a project budget of roughly \$10,000, and unblocked developers on hard technical problems.

Full-Stack Developer / Technical Project Manager | Freelance — Self-employed

Jun 2022 — Present

- Single point of accountability between client and delivery: requirements, architecture, build and handover.
- Led a team of 10 on an AI-enabled logistics system, ~\$13,000 budget over 11 months; built cross-platform Flutter apps, backend services and REST integrations.

Unreal Engine Developer / Independent Product Owner | Independent (indie)

Jan 2020 — Present

- Shipped 3 Android titles on Google Play (5,000+ combined downloads); built networked multiplayer with server authority and cloud player data.

Media Team Lead | Resala Charity Organization, Egypt

Jan 2020 — Dec 2020

- Led the media function at one of the largest charities in Egypt and the Middle East; completed a six-month leadership programme.

EDUCATION & CERTIFICATIONS

Bachelor — Computer Science & Engineering

Tomsk State University of Control Systems and Radioelectronics (TUSUR) · 2026

IBM Professional Course — Certificate of Appreciation · TUSUR — Recognition for contributing to an educational VR system · Unreal Engine course author (Udemy, 2024) · Digital Marketing (Google, 2021)

LANGUAGES

Arabic — Native · **English** — C2, full professional · **Russian** — B2, upper-intermediate